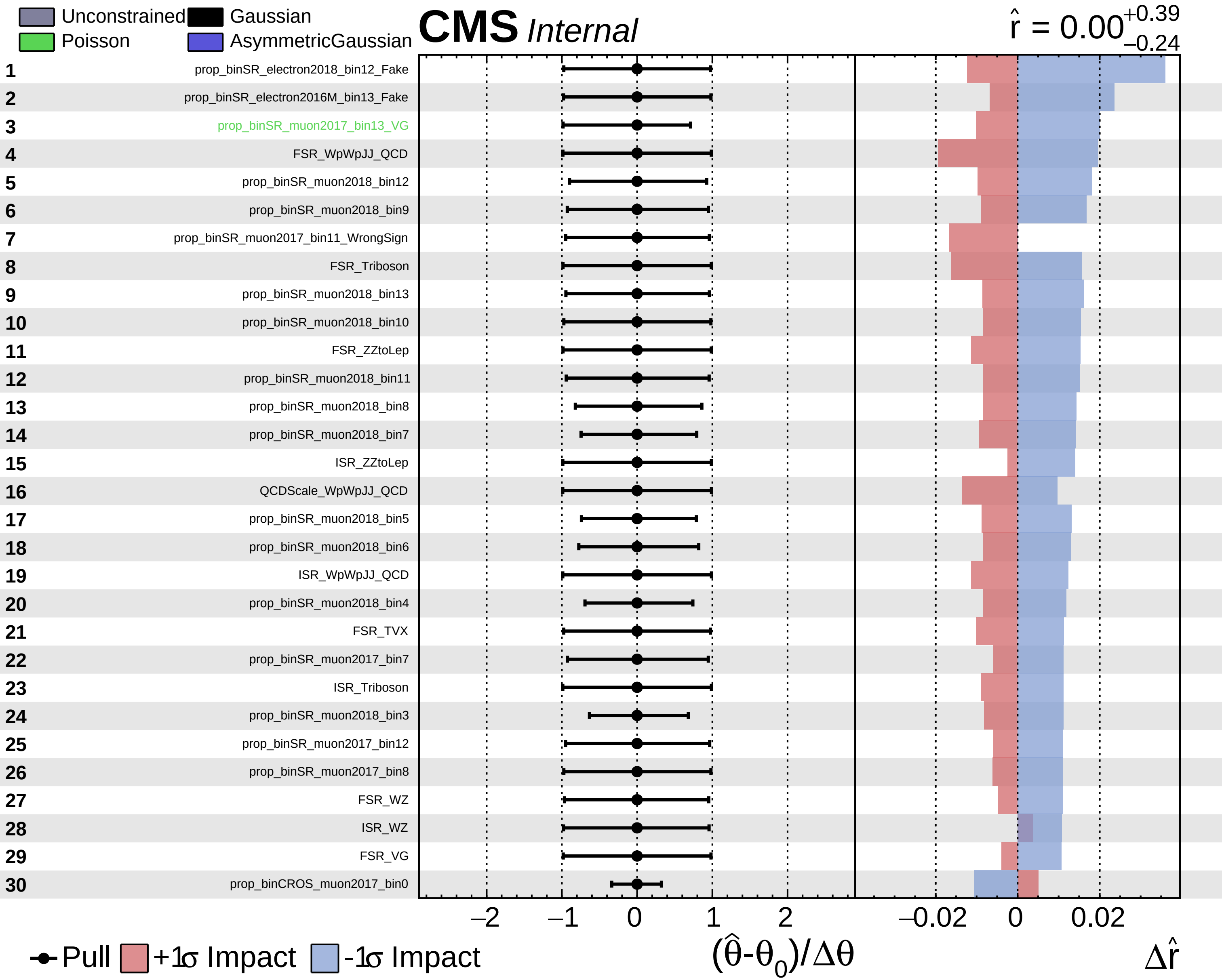
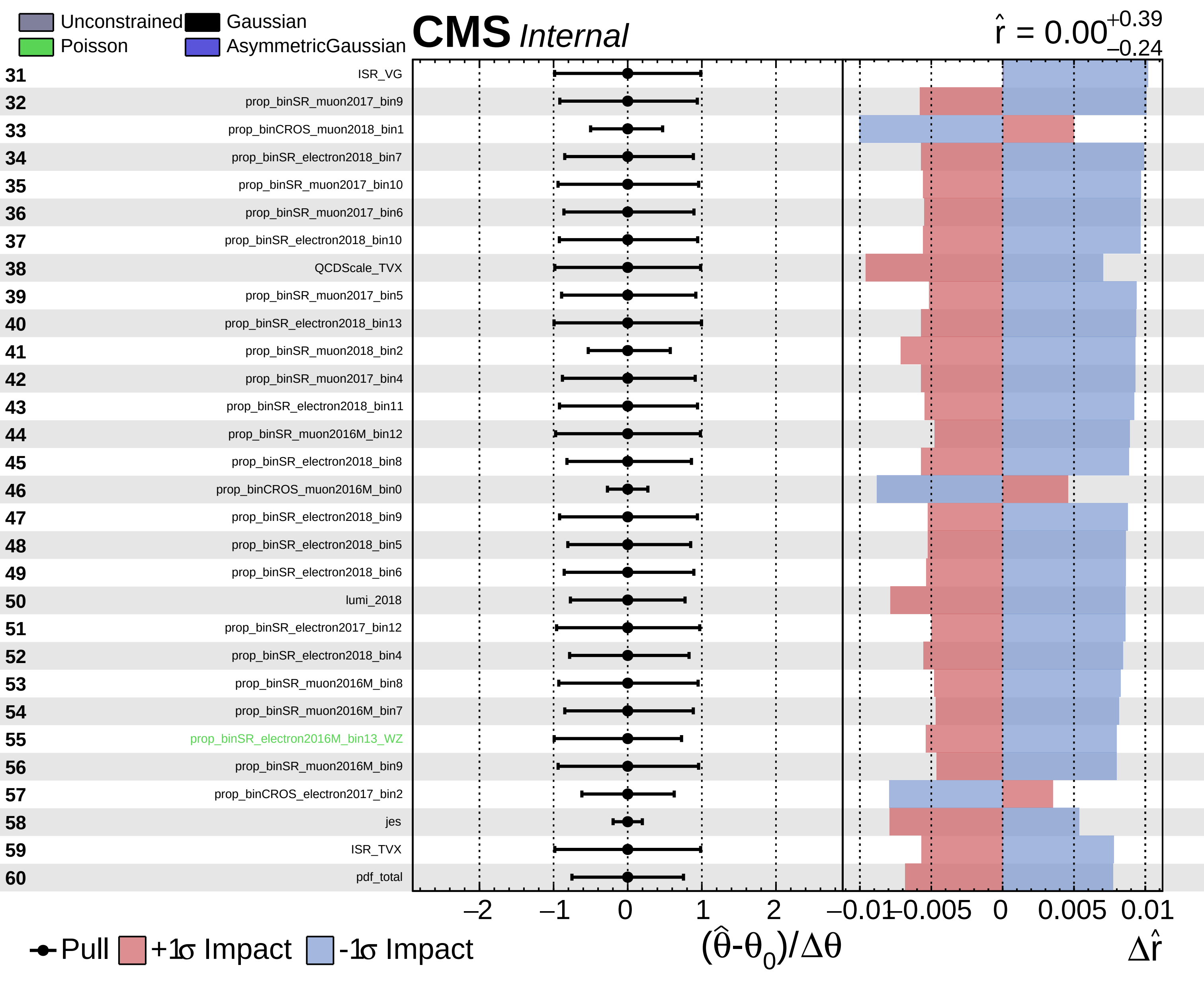
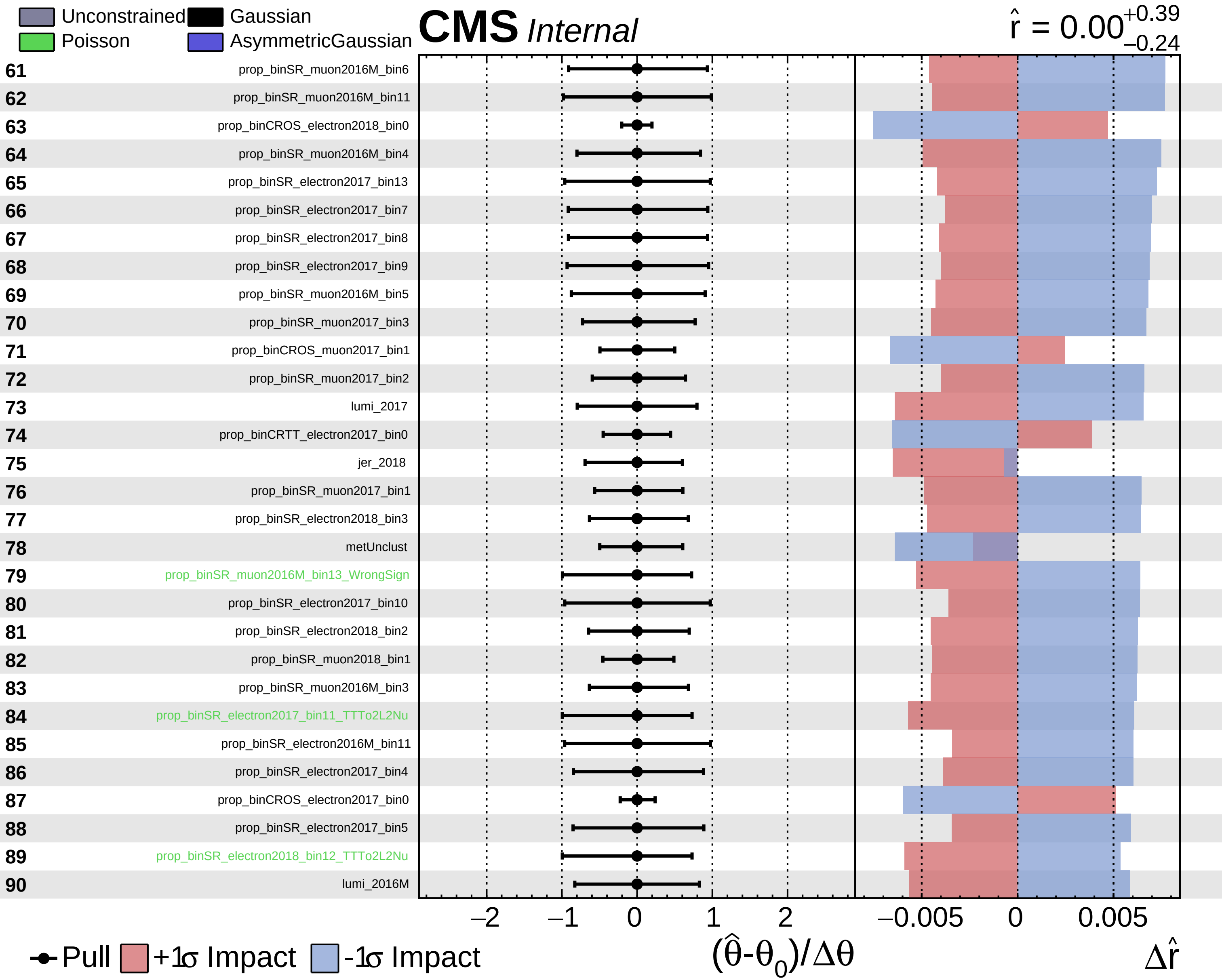
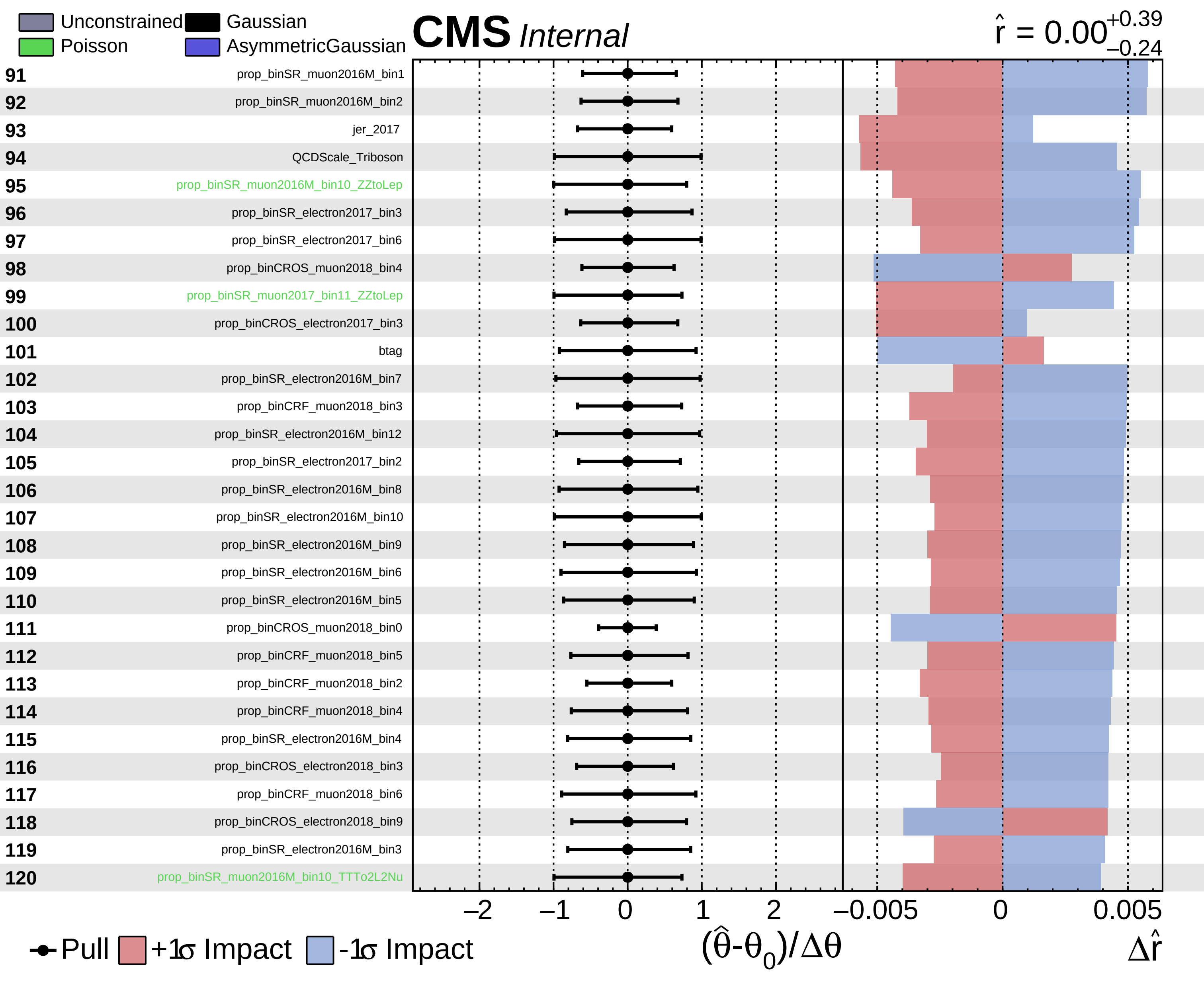


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
1 prop_binSR_electron2018_bin12_Fake
2 prop_binSR_electron2016M_bin13_Fake
3 prop_binSR_muon2017_bin13_VG
4 FSR_WpWpJJ_QCD
5 prop_binSR_muon2018_bin12
6 prop_binSR_muon2018_bin9
7 prop_binSR_muon2017_bin11_WrongSign
8 FSR_Triboson
9 prop_binSR_muon2018_bin13
10 prop_binSR_muon2018_bin10
11 FSR_ZZtoLep
12 prop_binSR_muon2018_bin11
13 prop_binSR_muon2018_bin8
14 prop_binSR_muon2018_bin7
15 ISR_ZZtoLep
16 QCDScale_WpWpJJ_QCD
17 prop_binSR_muon2018_bin5
18 prop_binSR_muon2018_bin6
19 ISR_WpWpJJ_QCD
20 prop_binSR_muon2018_bin4
21 FSR_TVX
22 prop_binSR_muon2017_bin7
23 ISR_Triboson
24 prop_binSR_muon2018_bin3
25 prop_binSR_muon2017_bin12
26 prop_binSR_muon2017_bin8
27 FSR_WZ
28 ISR_WZ
29 FSR_VG
30 prop_binCROS_muon2017_bin0
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

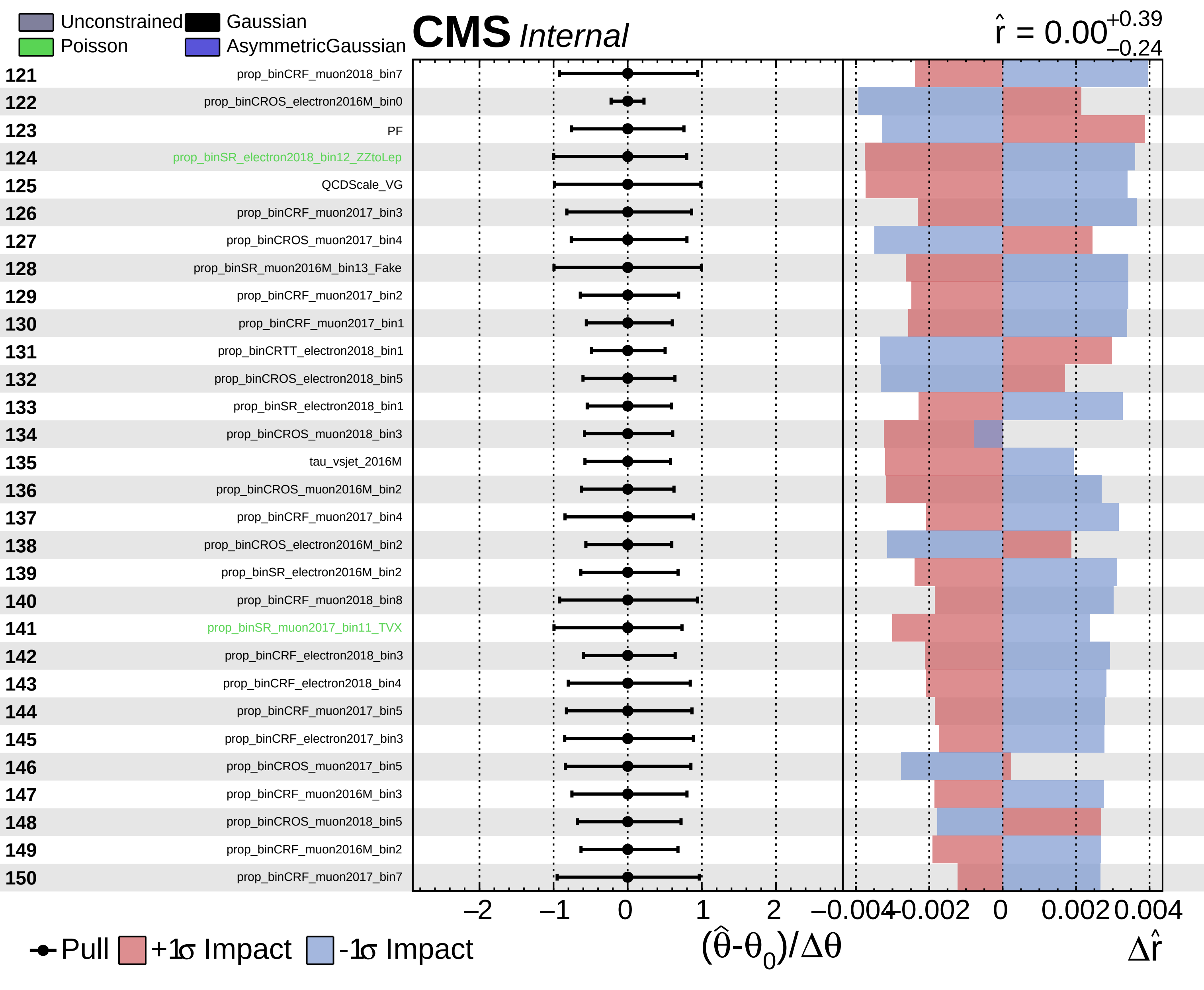

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
31 ISR_VG
32 prop_binSR_muon2017_bin9
33 prop_binCROS_muon2018_bin1
34 prop_binSR_electron2018_bin7
35 prop_binSR_muon2017_bin10
36 prop_binSR_muon2017_bin6
37 prop_binSR_electron2018_bin10
38 QCDScale_TVX
39 prop_binSR_muon2017_bin5
40 prop_binSR_electron2018_bin13
41 prop_binSR_muon2018_bin2
42 prop_binSR_muon2017_bin4
43 prop_binSR_electron2018_bin11
44 prop_binSR_muon2016M_bin12
45 prop_binSR_electron2018_bin8
46 prop_binCROS_muon2016M_bin0
47 prop_binSR_electron2018_bin9
48 prop_binSR_electron2018_bin5
49 prop_binSR_electron2018_bin6
50 lumi_2018
51 prop_binSR_electron2017_bin12
52 prop_binSR_electron2018_bin4
53 prop_binSR_muon2016M_bin8
54 prop_binSR_muon2016M_bin7
55 prop_binSR_electron2016M_bin13_WZ
56 prop_binSR_muon2016M_bin9
57 prop_binCROS_electron2017_bin2
58 jes
59 ISR_TVX
60 pdf_total
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



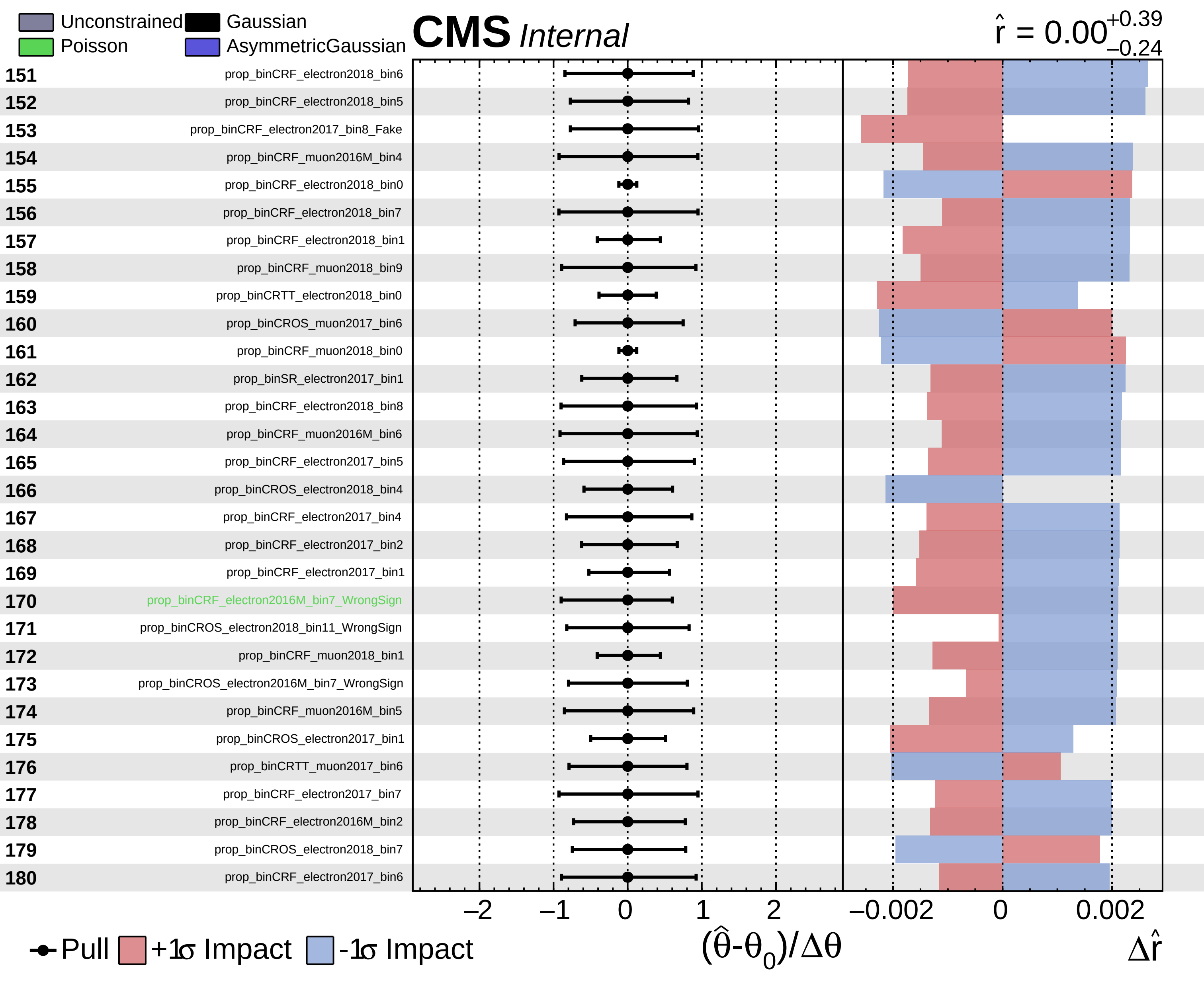
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
61 prop_binSR_muon2016M_bin6
62 prop_binSR_muon2016M_bin11
63 prop_binCROS_electron2018_bin0
64 prop_binSR_muon2016M_bin4
65 prop_binSR_electron2017_bin13
66 prop_binSR_electron2017_bin7
67 prop_binSR_electron2017_bin8
68 prop_binSR_electron2017_bin9
69 prop_binSR_muon2016M_bin5
70 prop_binSR_muon2017_bin3
71 prop_binCROS_muon2017_bin1
72 prop_binSR_muon2017_bin2
73 lumi_2017
74 prop_binCRTT_electron2017_bin0
75 jer_2018
76 prop_binSR_muon2017_bin1
77 prop_binSR_electron2018_bin3
78 metUnclust
79 prop_binSR_muon2016M_bin13_WrongSign
80 prop_binSR_electron2017_bin10
81 prop_binSR_electron2018_bin2
82 prop_binSR_muon2018_bin1
83 prop_binSR_muon2016M_bin3
84 prop_binSR_electron2017_bin11_TTTo2L2Nu
85 prop_binSR_electron2016M_bin11
86 prop_binSR_electron2017_bin4
87 prop_binCROS_electron2017_bin0
88 prop_binSR_electron2017_bin5
89 prop_binSR_electron2018_bin12_TTTo2L2Nu
90 lumi_2016M
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



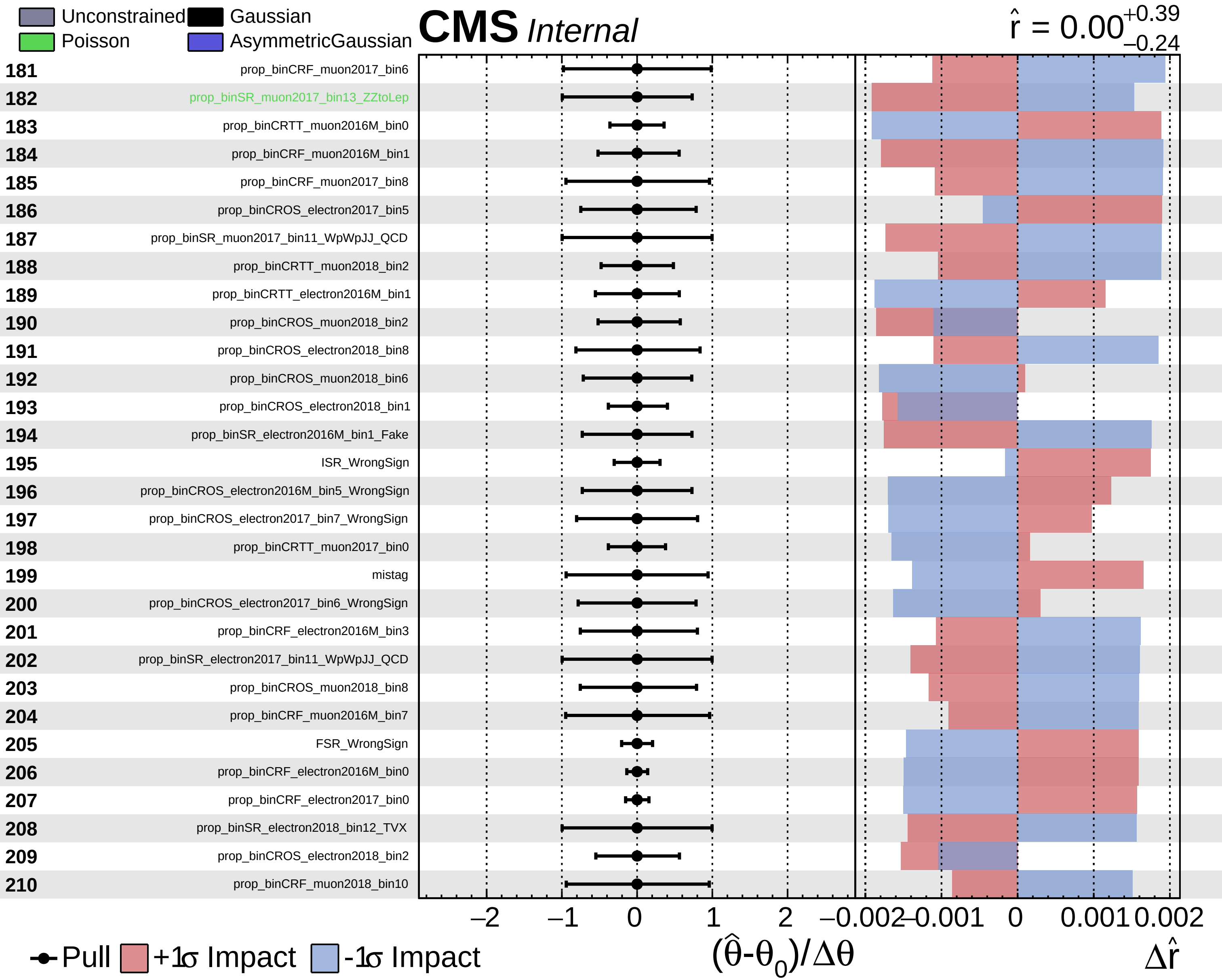
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
91 prop_binSR_muon2016M_bin1
92 prop_binSR_muon2016M_bin2
93 jer_2017
94 QCDScale_Triboson
95 prop_binSR_muon2016M_bin10_ZZtoLep
96 prop_binSR_electron2017_bin3
97 prop_binSR_electron2017_bin6
98 prop_binCROS_muon2018_bin4
99 prop_binSR_muon2017_bin11_ZZtoLep
100 prop_binCROS_electron2017_bin3
101 btag
102 prop_binSR_electron2016M_bin7
103 prop_binCRF_muon2018_bin3
104 prop_binSR_electron2016M_bin12
105 prop_binSR_electron2017_bin2
106 prop_binSR_electron2016M_bin8
107 prop_binSR_electron2016M_bin10
108 prop_binSR_electron2016M_bin9
109 prop_binSR_electron2016M_bin6
110 prop_binSR_electron2016M_bin5
111 prop_binCROS_muon2018_bin0
112 prop_binCRF_muon2018_bin5
113 prop_binCRF_muon2018_bin2
114 prop_binCRF_muon2018_bin4
115 prop_binSR_electron2016M_bin4
116 prop_binCROS_electron2018_bin3
117 prop_binCRF_muon2018_bin6
118 prop_binCROS_electron2018_bin9
119 prop_binSR_electron2016M_bin3
120 prop_binSR_muon2016M_bin10_TTTo2L2Nu
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



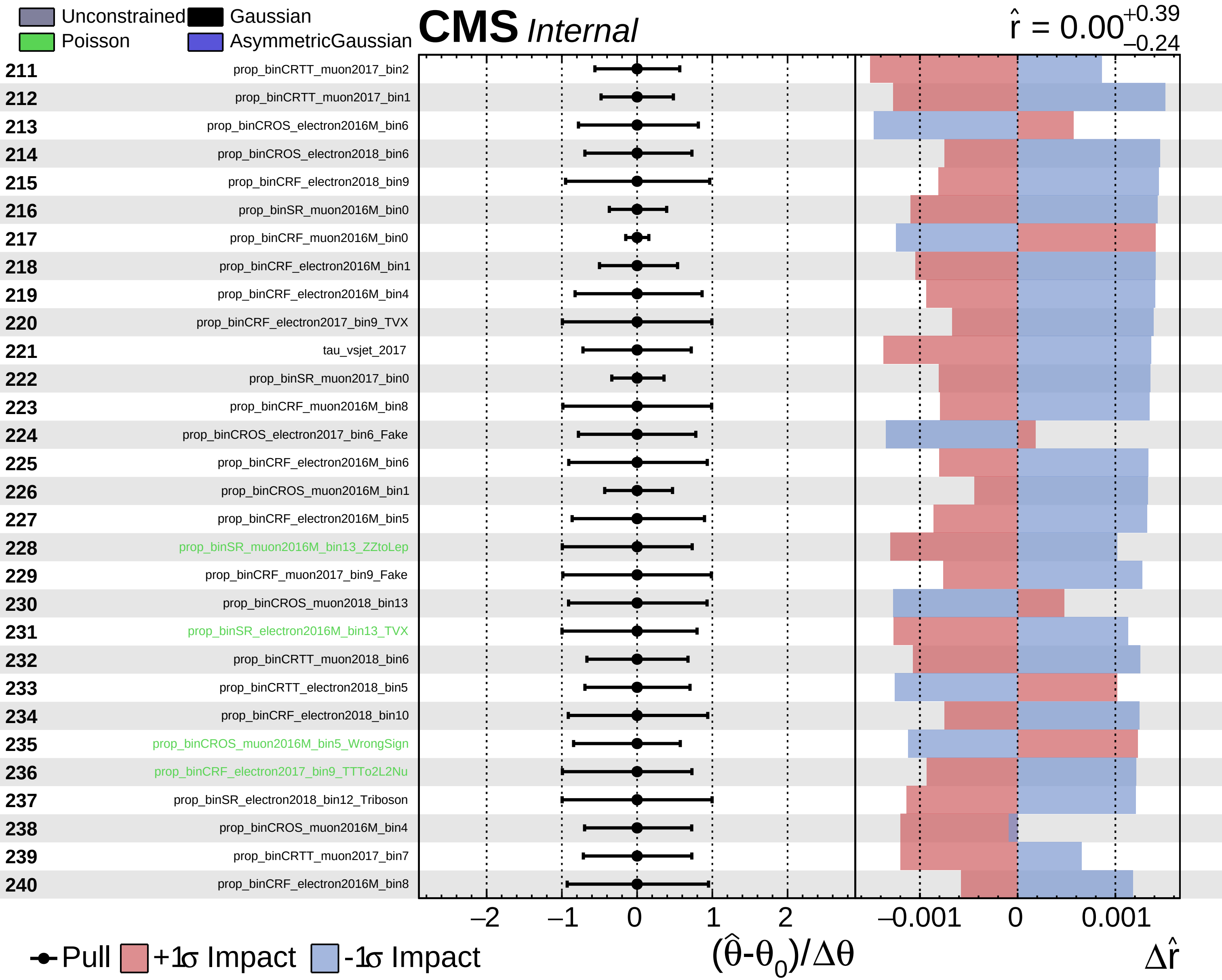
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
121 prop_binCRF_muon2018_bin7
122 prop_binCROS_electron2016M_bin0
123 PF
124 prop_binSR_electron2018_bin12_ZZtoLep
125 QCDScale_VG
126 prop_binCRF_muon2017_bin3
127 prop_binCROS_muon2017_bin4
128 prop_binSR_muon2016M_bin13_Fake
129 prop_binCRF_muon2017_bin2
130 prop_binCRF_muon2017_bin1
131 prop_binCRTT_electron2018_bin1
132 prop_binCROS_electron2018_bin5
133 prop_binSR_electron2018_bin1
134 prop_binCROS_muon2018_bin3
135 tau_vsjet_2016M
136 prop_binCROS_muon2016M_bin2
137 prop_binCRF_muon2017_bin4
138 prop_binCROS_electron2016M_bin2
139 prop_binSR_electron2016M_bin2
140 prop_binCRF_muon2018_bin8
141 prop_binSR_muon2017_bin11_TVX
142 prop_binCRF_electron2018_bin3
143 prop_binCRF_electron2018_bin4
144 prop_binCRF_muon2017_bin5
145 prop_binCRF_electron2017_bin3
146 prop_binCROS_muon2017_bin5
147 prop_binCRF_muon2016M_bin3
148 prop_binCROS_muon2018_bin5
149 prop_binCRF_muon2016M_bin2
150 prop_binCRF_muon2017_bin7
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



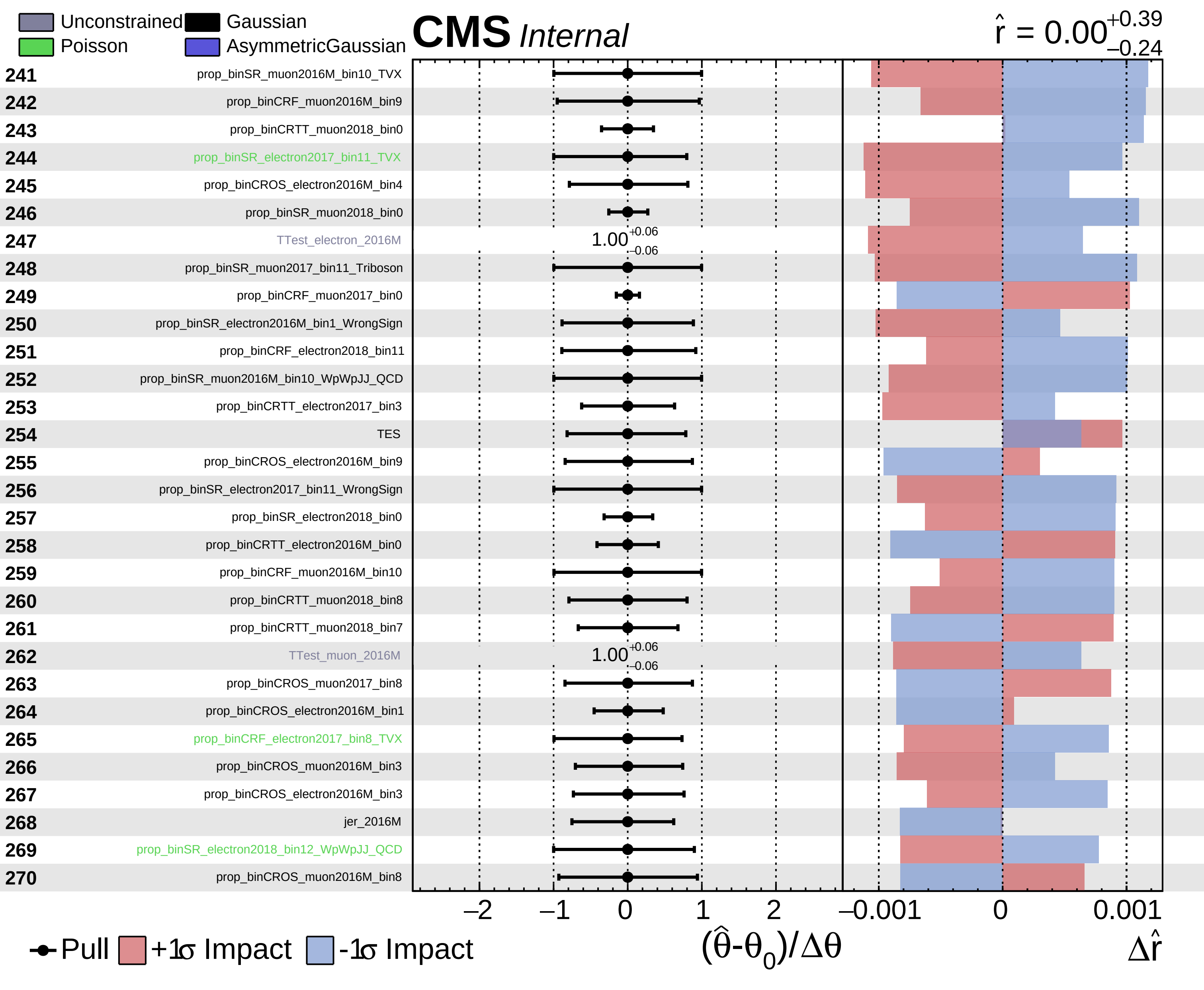
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
151 prop_binCRF_electron2018_bin6
152 prop_binCRF_electron2018_bin5
153 prop_binCRF_electron2017_bin8_Fake
154 prop_binCRF_muon2016M_bin4
155 prop_binCRF_electron2018_bin0
156 prop_binCRF_electron2018_bin7
157 prop_binCRF_electron2018_bin1
158 prop_binCRF_muon2018_bin9
159 prop_binCRTT_electron2018_bin0
160 prop_binCROS_muon2017_bin6
161 prop_binCRF_muon2018_bin0
162 prop_binSR_electron2017_bin1
163 prop_binCRF_electron2018_bin8
164 prop_binCRF_muon2016M_bin6
165 prop_binCRF_electron2017_bin5
166 prop_binCROS_electron2018_bin4
167 prop_binCRF_electron2017_bin4
168 prop_binCRF_electron2017_bin2
169 prop_binCRF_electron2017_bin1
170 prop_binCRF_electron2016M_bin7_WrongSign
171 prop_binCROS_electron2018_bin11_WrongSign
172 prop_binCRF_muon2018_bin1
173 prop_binCROS_electron2016M_bin7_WrongSign
174 prop_binCRF_muon2016M_bin5
175 prop_binCROS_electron2017_bin1
176 prop_binCRTT_muon2017_bin6
177 prop_binCRF_electron2017_bin7
178 prop_binCRF_electron2016M_bin2
179 prop_binCROS_electron2018_bin7
180 prop_binCRF_electron2017_bin6
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



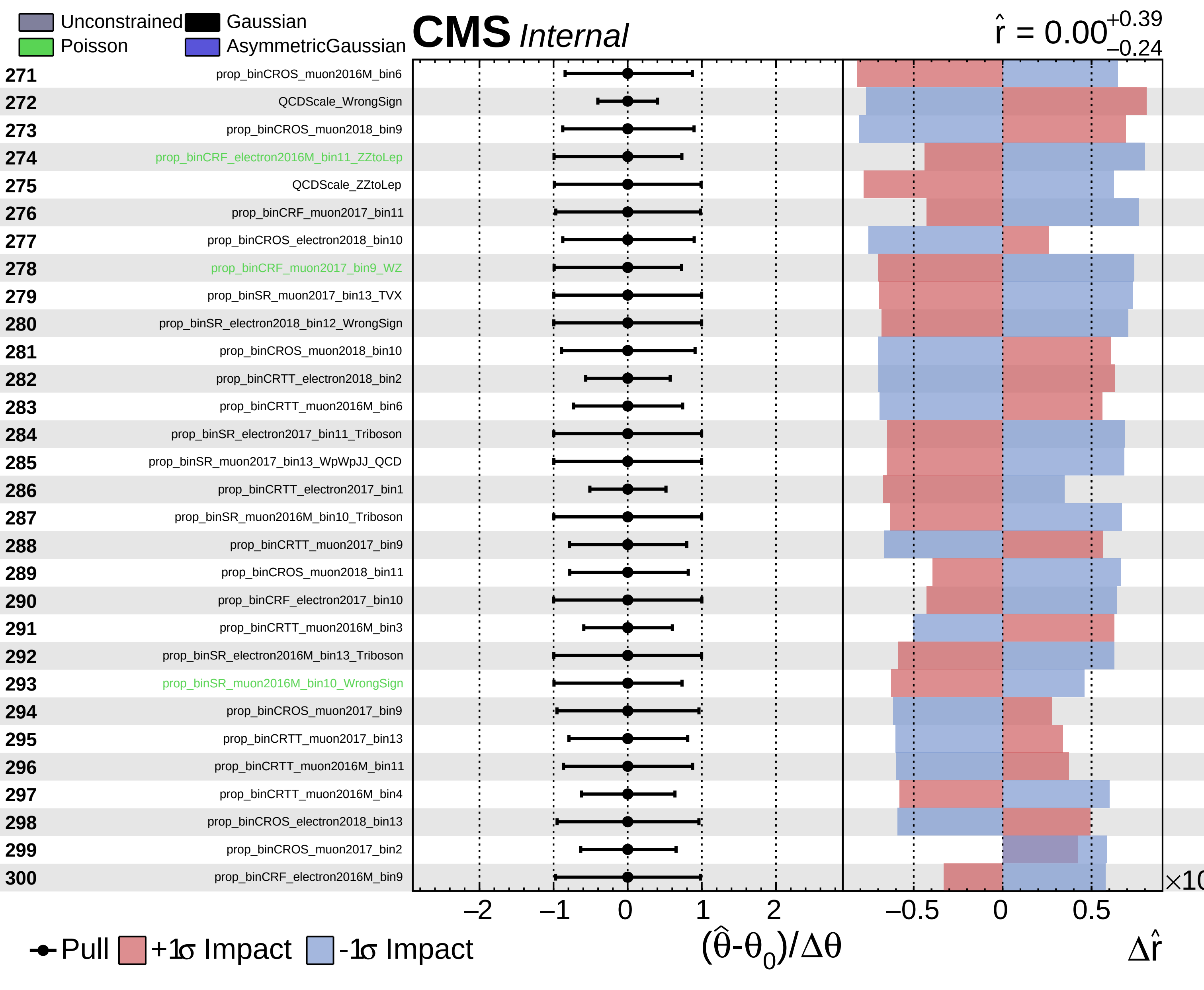
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
181 prop_binCRF_muon2017_bin6
182 prop_binSR_muon2017_bin13_ZZtoLep
183 prop_binCRTT_muon2016M_bin0
184 prop_binCRF_muon2016M_bin1
185 prop_binCRF_muon2017_bin8
186 prop_binCROS_electron2017_bin5
187 prop_binSR_muon2017_bin11_WpWpJJ_QCD
188 prop_binCRTT_muon2018_bin2
189 prop_binCRTT_electron2016M_bin1
190 prop_binCROS_muon2018_bin2
191 prop_binCROS_electron2018_bin8
192 prop_binCROS_muon2018_bin6
193 prop_binCROS_electron2018_bin1
194 prop_binSR_electron2016M_bin1_Fake
195 ISR_WrongSign
196 prop_binCROS_electron2016M_bin5_WrongSign
197 prop_binCROS_electron2017_bin7_WrongSign
198 prop_binCRTT_muon2017_bin0
199 mistag
200 prop_binCROS_electron2017_bin6_WrongSign
201 prop_binCRF_electron2016M_bin3
202 prop_binSR_electron2017_bin11_WpWpJJ_QCD
203 prop_binCROS_muon2018_bin8
204 prop_binCRF_muon2016M_bin7
205 FSR_WrongSign
206 prop_binCRF_electron2016M_bin0
207 prop_binCRF_electron2017_bin0
208 prop_binSR_electron2018_bin12_TVX
209 prop_binCROS_electron2018_bin2
210 prop_binCRF_muon2018_bin10
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



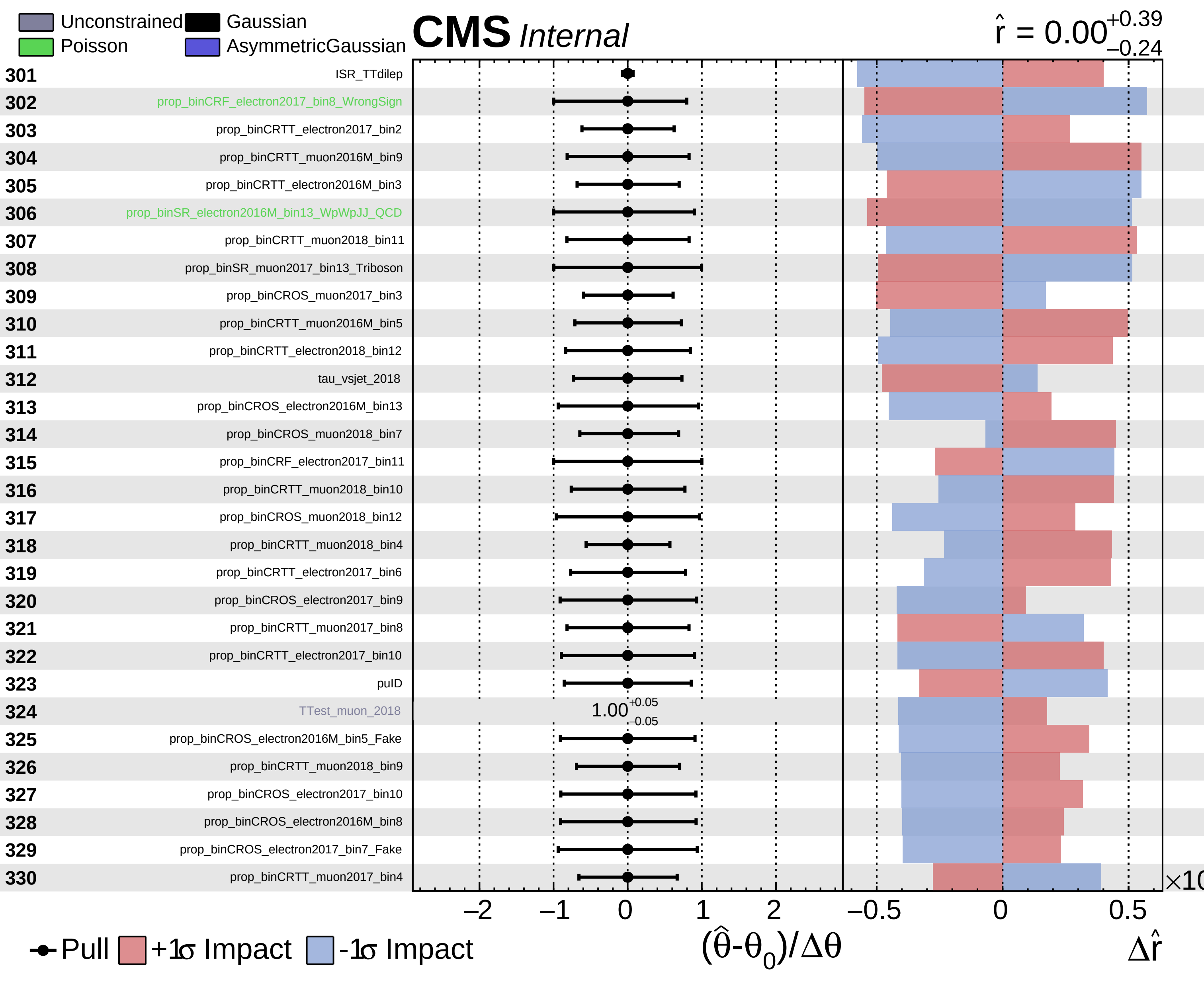
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
211 prop_binCRTT_muon2017_bin2
212 prop_binCRTT_muon2017_bin1
213 prop_binCROS_electron2016M_bin6
214 prop_binCROS_electron2018_bin6
215 prop_binCRF_electron2018_bin9
216 prop_binSR_muon2016M_bin0
217 prop_binCRF_muon2016M_bin0
218 prop_binCRF_electron2016M_bin1
219 prop_binCRF_electron2016M_bin4
220 prop_binCRF_electron2017_bin9_TVX
221 tau_vsjet_2017
222 prop_binSR_muon2017_bin0
223 prop_binCRF_muon2016M_bin8
224 prop_binCROS_electron2017_bin6_Fake
225 prop_binCRF_electron2016M_bin6
226 prop_binCROS_muon2016M_bin1
227 prop_binCRF_electron2016M_bin5
228 prop_binSR_muon2016M_bin13_ZZtoLep
229 prop_binCRF_muon2017_bin9_Fake
230 prop_binCROS_muon2018_bin13
231 prop_binSR_electron2016M_bin13_TVX
232 prop_binCRTT_muon2018_bin6
233 prop_binCRTT_electron2018_bin5
234 prop_binCRF_electron2018_bin10
235 prop_binCROS_muon2016M_bin5_WrongSign
236 prop_binCRF_electron2017_bin9_TTTo2L2Nu
237 prop_binSR_electron2018_bin12_Triboson
238 prop_binCROS_muon2016M_bin4
239 prop_binCRTT_muon2017_bin7
240 prop_binCRF_electron2016M_bin8
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



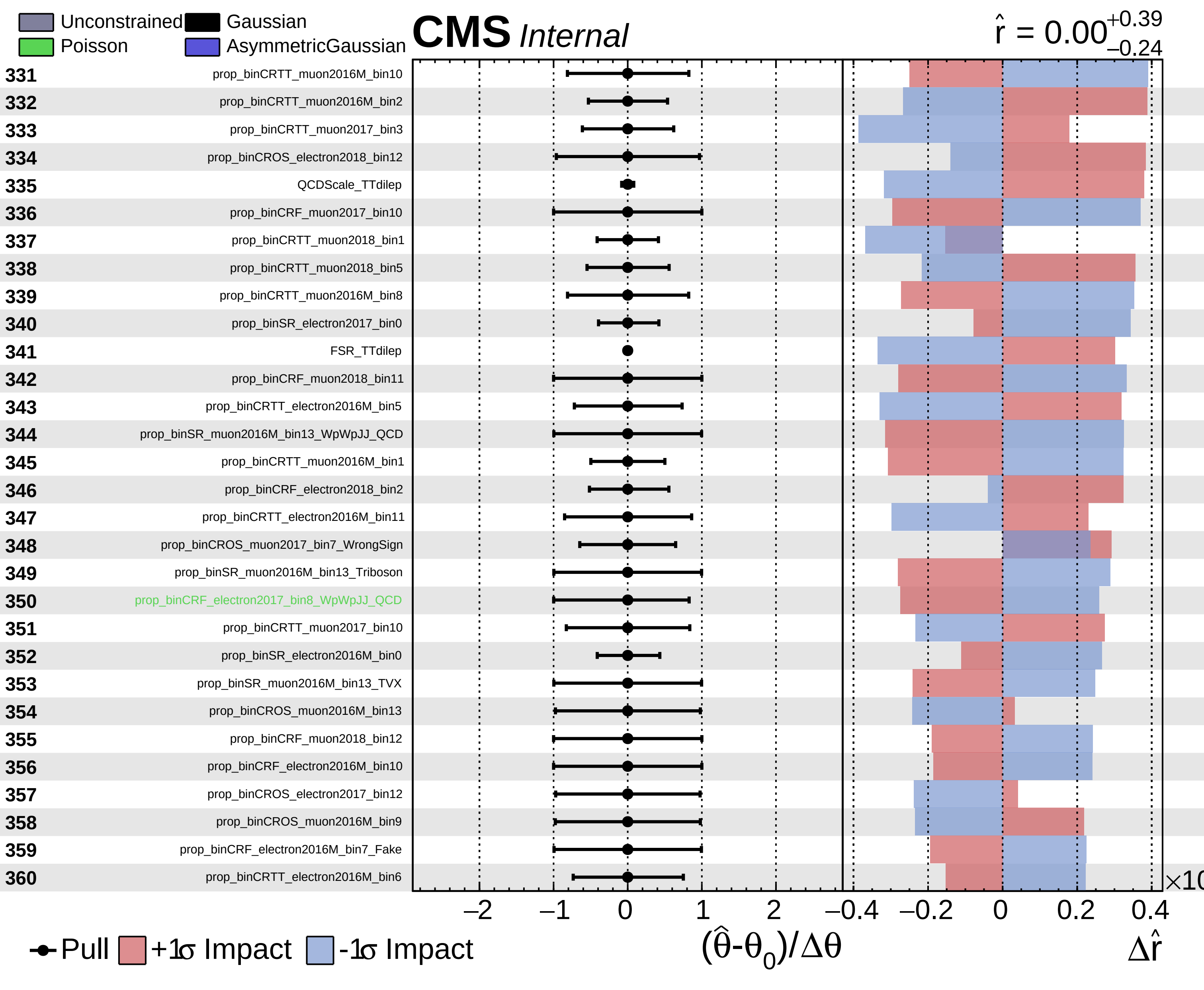
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.39 −0.24
241 prop_binSR_muon2016M_bin10_TVX
242 prop_binCRF_muon2016M_bin9
243 prop_binCRTT_muon2018_bin0
244 prop_binSR_electron2017_bin11_TVX
245 prop_binCROS_electron2016M_bin4
246 prop_binSR_muon2018_bin0
247 TTest_electron_2016M
1.00 +0.06 −0.06
248 prop_binSR_muon2017_bin11_Triboson
249 prop_binCRF_muon2017_bin0
250 prop_binSR_electron2016M_bin1_WrongSign
251 prop_binCRF_electron2018_bin11
252 prop_binSR_muon2016M_bin10_WpWpJJ_QCD
253 prop_binCRTT_electron2017_bin3
254 TES
255 prop_binCROS_electron2016M_bin9
256 prop_binSR_electron2017_bin11_WrongSign
257 prop_binSR_electron2018_bin0
258 prop_binCRTT_electron2016M_bin0
259 prop_binCRF_muon2016M_bin10
260 prop_binCRTT_muon2018_bin8
261 prop_binCRTT_muon2018_bin7
262 TTest_muon_2016M
1.00 +0.06 −0.06
263 prop_binCROS_muon2017_bin8
264 prop_binCROS_electron2016M_bin1
265 prop_binCRF_electron2017_bin8_TVX
266 prop_binCROS_muon2016M_bin3
267 prop_binCROS_electron2016M_bin3
268 jer_2016M
269 prop_binSR_electron2018_bin12_WpWpJJ_QCD
270 prop_binCROS_muon2016M_bin8
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

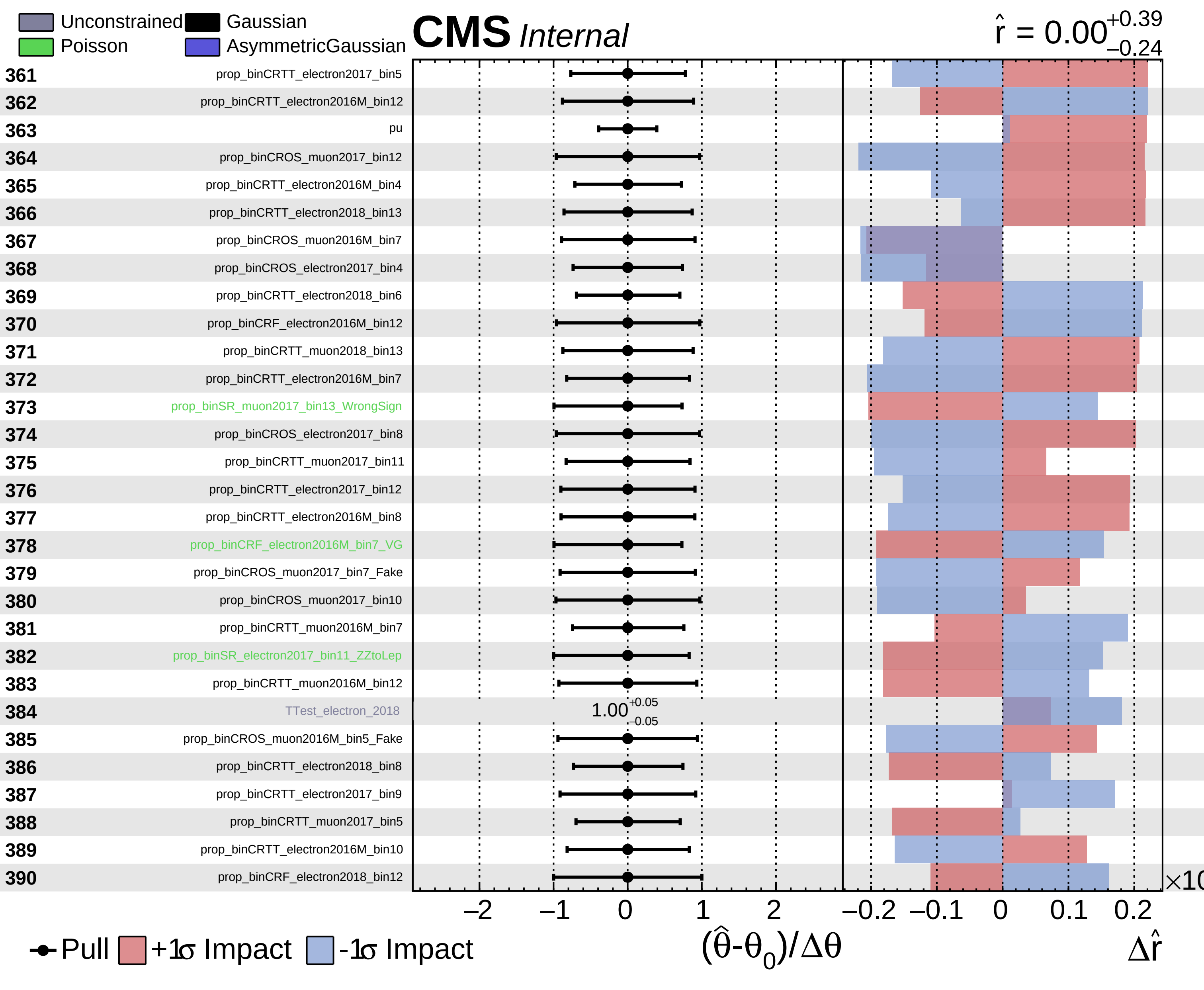


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
271 prop_binCROS_muon2016M_bin6
272 QCDScale_WrongSign
273 prop_binCROS_muon2018_bin9
274 prop_binCRF_electron2016M_bin11_ZZtoLep
275 QCDScale_ZZtoLep
276 prop_binCRF_muon2017_bin11
277 prop_binCROS_electron2018_bin10
278 prop_binCRF_muon2017_bin9_WZ
279 prop_binSR_muon2017_bin13_TVX
280 prop_binSR_electron2018_bin12_WrongSign
281 prop_binCROS_muon2018_bin10
282 prop_binCRTT_electron2018_bin2
283 prop_binCRTT_muon2016M_bin6
284 prop_binSR_electron2017_bin11_Triboson
285 prop_binSR_muon2017_bin13_WpWpJJ_QCD
286 prop_binCRTT_electron2017_bin1
287 prop_binSR_muon2016M_bin10_Triboson
288 prop_binCRTT_muon2017_bin9
289 prop_binCROS_muon2018_bin11
290 prop_binCRF_electron2017_bin10
291 prop_binCRTT_muon2016M_bin3
292 prop_binSR_electron2016M_bin13_Triboson
293 prop_binSR_muon2016M_bin10_WrongSign
294 prop_binCROS_muon2017_bin9
295 prop_binCRTT_muon2017_bin13
296 prop_binCRTT_muon2016M_bin11
297 prop_binCRTT_muon2016M_bin4
298 prop_binCROS_electron2018_bin13
299 prop_binCROS_muon2017_bin2
300 prop_binCRF_electron2016M_bin9
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

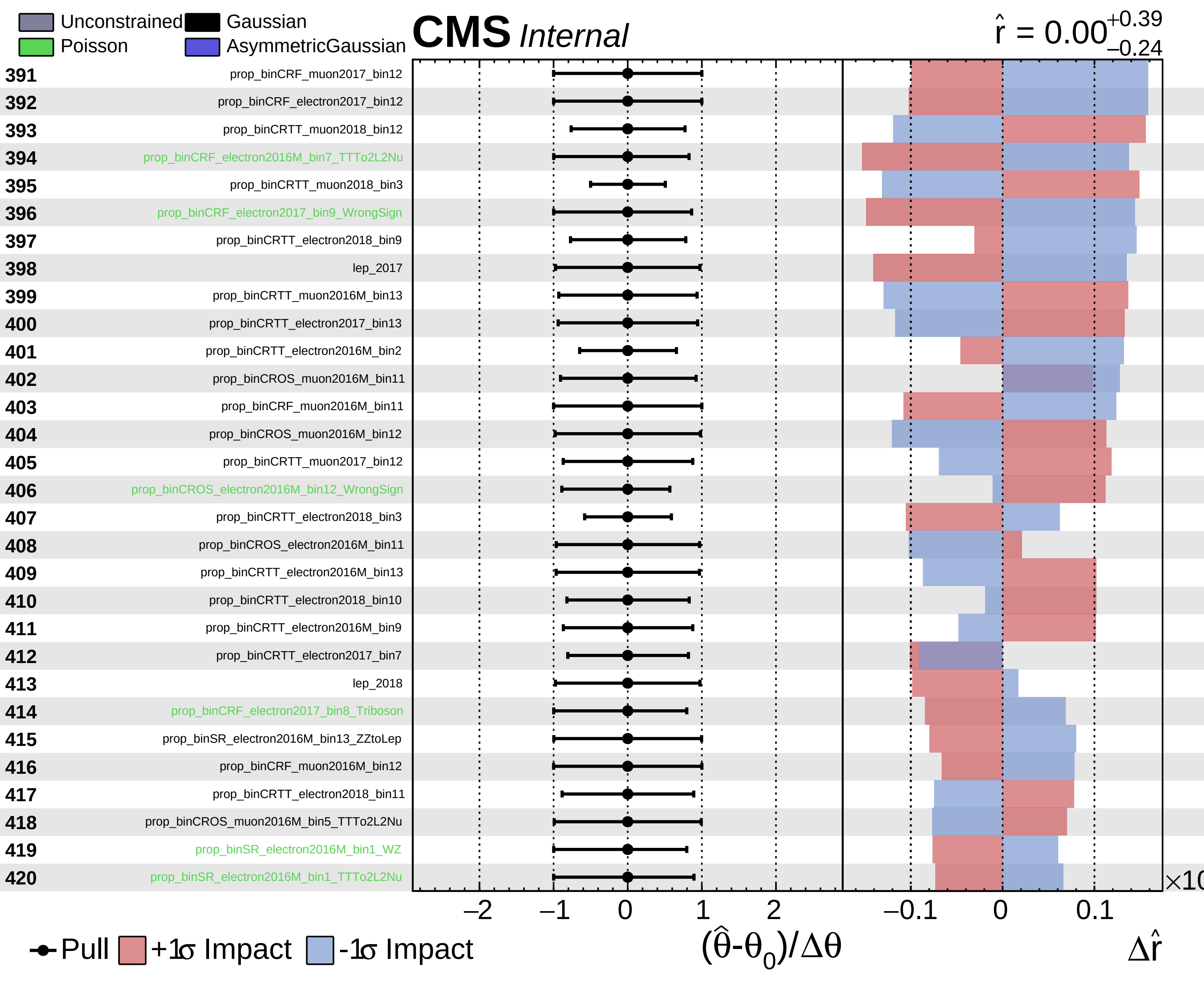


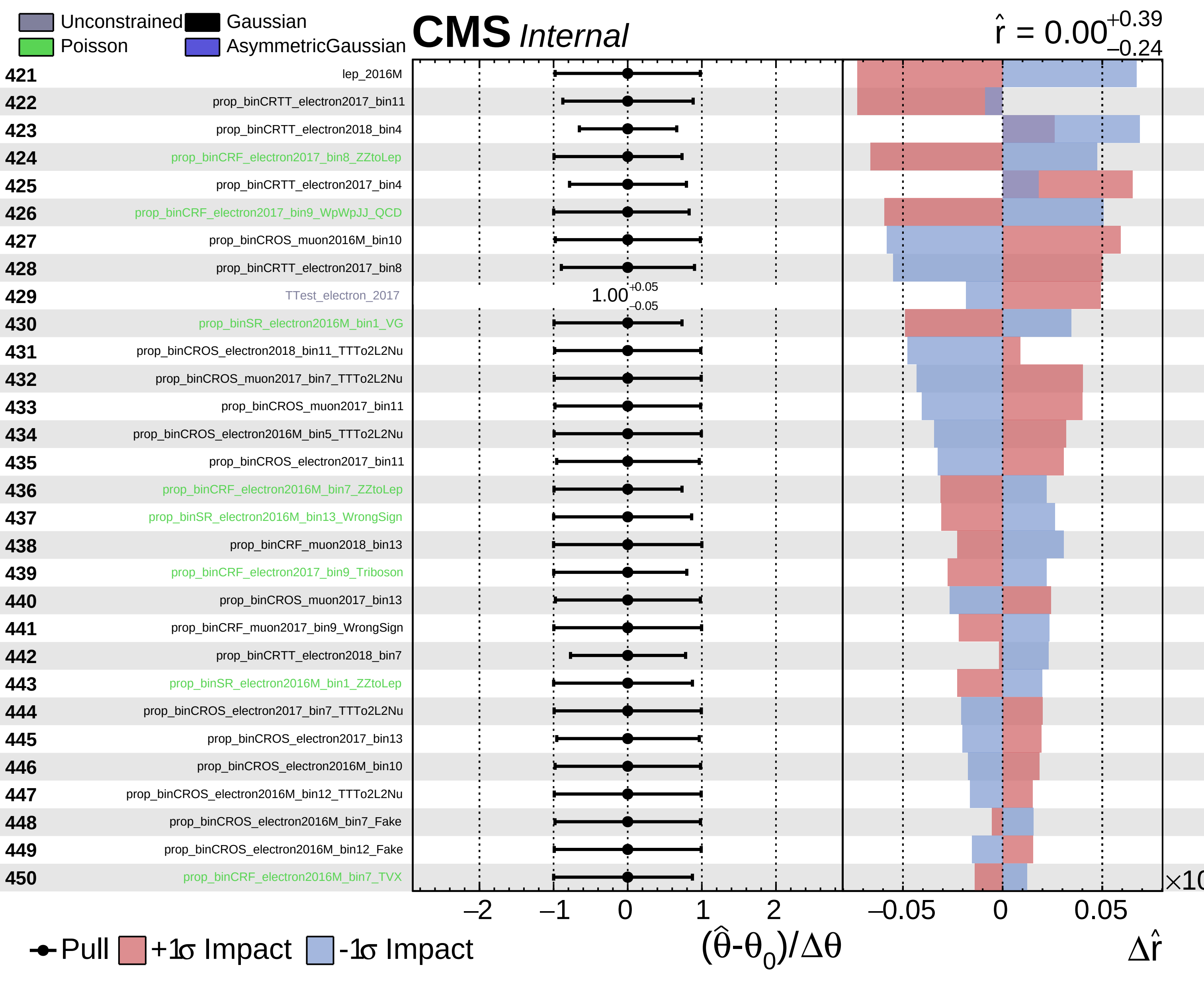
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
301 ISR_TTdilep
302 prop_binCRF_electron2017_bin8_WrongSign
303 prop_binCRTT_electron2017_bin2
304 prop_binCRTT_muon2016M_bin9
305 prop_binCRTT_electron2016M_bin3
306 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
307 prop_binCRTT_muon2018_bin11
308 prop_binSR_muon2017_bin13_Triboson
309 prop_binCROS_muon2017_bin3
310 prop_binCRTT_muon2016M_bin5
311 prop_binCRTT_electron2018_bin12
312 tau_vsjet_2018
313 prop_binCROS_electron2016M_bin13
314 prop_binCROS_muon2018_bin7
315 prop_binCRF_electron2017_bin11
316 prop_binCRTT_muon2018_bin10
317 prop_binCROS_muon2018_bin12
318 prop_binCRTT_muon2018_bin4
319 prop_binCRTT_electron2017_bin6
320 prop_binCROS_electron2017_bin9
321 prop_binCRTT_muon2017_bin8
322 prop_binCRTT_electron2017_bin10
323 puID
324 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
325 prop_binCROS_electron2016M_bin5_Fake
326 prop_binCRTT_muon2018_bin9
327 prop_binCROS_electron2017_bin10
328 prop_binCROS_electron2016M_bin8
329 prop_binCROS_electron2017_bin7_Fake
330 prop_binCRTT_muon2017_bin4
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
331 prop_binCRTT_muon2016M_bin10
332 prop_binCRTT_muon2016M_bin2
333 prop_binCRTT_muon2017_bin3
334 prop_binCROS_electron2018_bin12
335 QCDScale_TTdilep
336 prop_binCRF_muon2017_bin10
337 prop_binCRTT_muon2018_bin1
338 prop_binCRTT_muon2018_bin5
339 prop_binCRTT_muon2016M_bin8
340 prop_binSR_electron2017_bin0
341 FSR_TTdilep
342 prop_binCRF_muon2018_bin11
343 prop_binCRTT_electron2016M_bin5
344 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
345 prop_binCRTT_muon2016M_bin1
346 prop_binCRF_electron2018_bin2
347 prop_binCRTT_electron2016M_bin11
348 prop_binCROS_muon2017_bin7_WrongSign
349 prop_binSR_muon2016M_bin13_Triboson
350 prop_binCRF_electron2017_bin8_WpWpJJ_QCD
351 prop_binCRTT_muon2017_bin10
352 prop_binSR_electron2016M_bin0
353 prop_binSR_muon2016M_bin13_TVX
354 prop_binCROS_muon2016M_bin13
355 prop_binCRF_muon2018_bin12
356 prop_binCRF_electron2016M_bin10
357 prop_binCROS_electron2017_bin12
358 prop_binCROS_muon2016M_bin9
359 prop_binCRF_electron2016M_bin7_Fake
360 prop_binCRTT_electron2016M_bin6
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

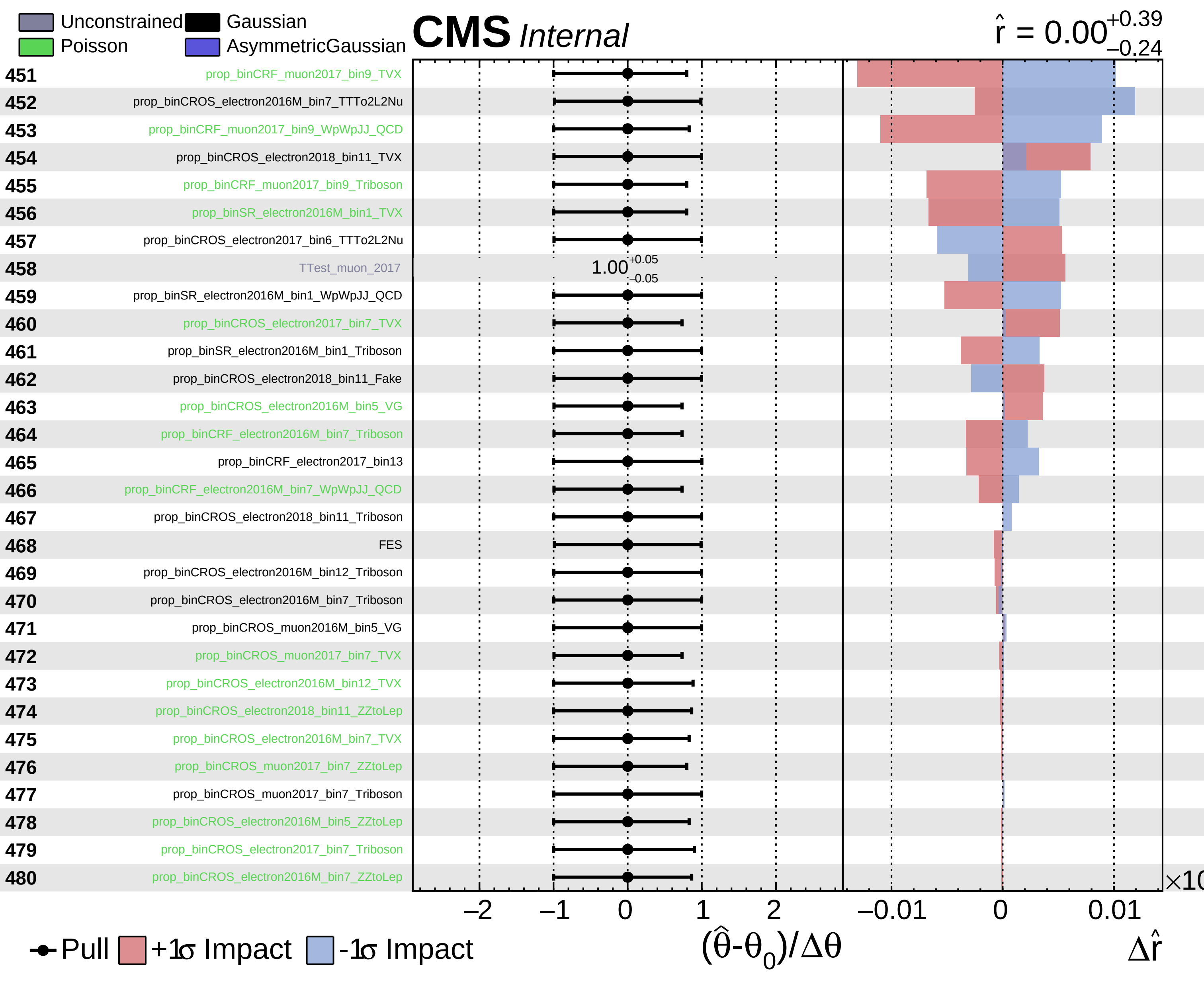


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
361 prop_binCRTT_electron2017_bin5
362 prop_binCRTT_electron2016M_bin12
363 pu
364 prop_binCROS_muon2017_bin12
365 prop_binCRTT_electron2016M_bin4
366 prop_binCRTT_electron2018_bin13
367 prop_binCROS_muon2016M_bin7
368 prop_binCROS_electron2017_bin4
369 prop_binCRTT_electron2018_bin6
370 prop_binCRF_electron2016M_bin12
371 prop_binCRTT_muon2018_bin13
372 prop_binCRTT_electron2016M_bin7
373 prop_binSR_muon2017_bin13_WrongSign
374 prop_binCROS_electron2017_bin8
375 prop_binCRTT_muon2017_bin11
376 prop_binCRTT_electron2017_bin12
377 prop_binCRTT_electron2016M_bin8
378 prop_binCRF_electron2016M_bin7_VG
379 prop_binCROS_muon2017_bin7_Fake
380 prop_binCROS_muon2017_bin10
381 prop_binCRTT_muon2016M_bin7
382 prop_binSR_electron2017_bin11_ZZtoLep
383 prop_binCRTT_muon2016M_bin12
384 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
385 prop_binCROS_muon2016M_bin5_Fake
386 prop_binCRTT_electron2018_bin8
387 prop_binCRTT_electron2017_bin9
388 prop_binCRTT_muon2017_bin5
389 prop_binCRTT_electron2016M_bin10
390 prop_binCRF_electron2018_bin12
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

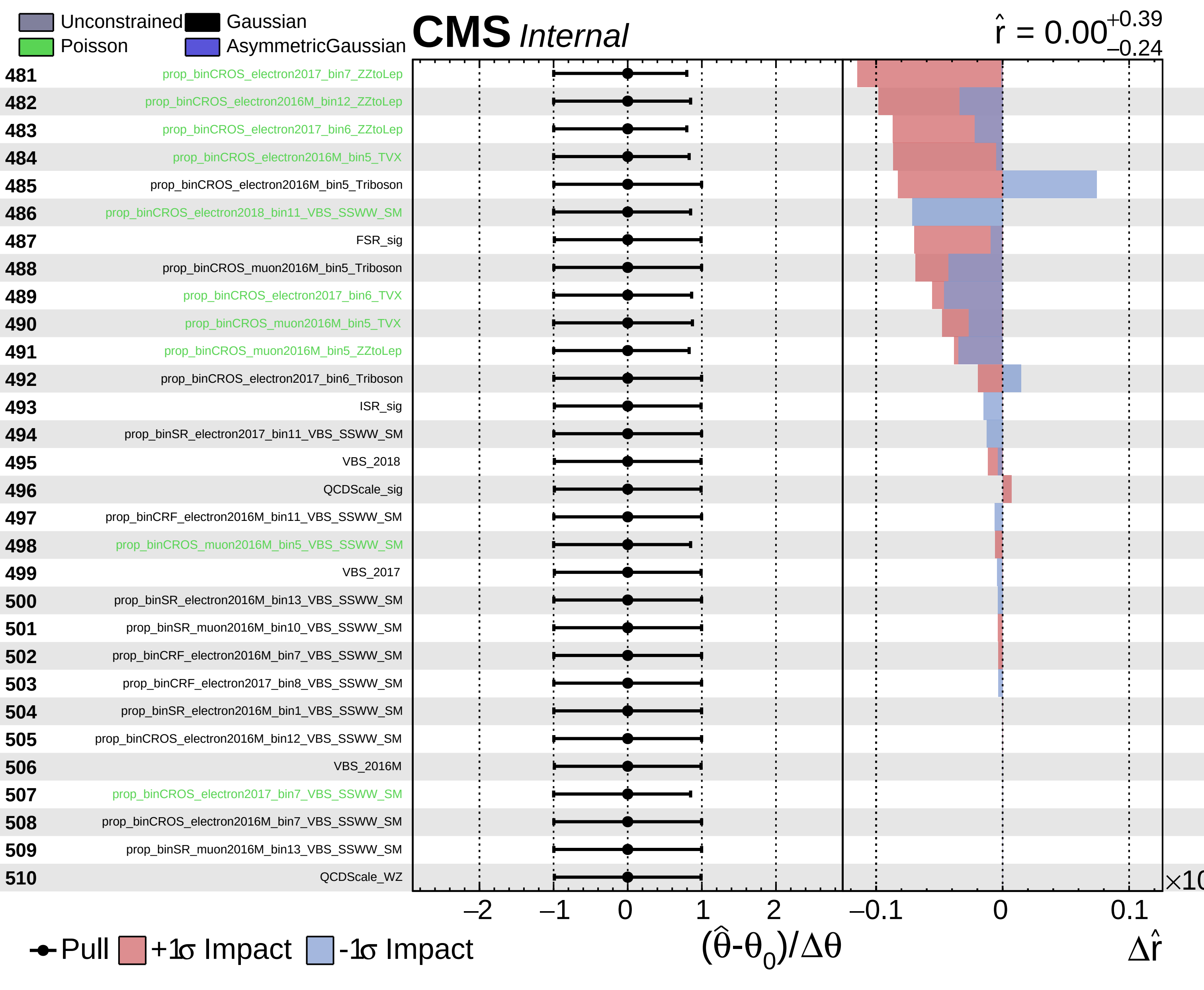

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.39 −0.24
391 prop_binCRF_muon2017_bin12
392 prop_binCRF_electron2017_bin12
393 prop_binCRTT_muon2018_bin12
394 prop_binCRF_electron2016M_bin7_TTTo2L2Nu
395 prop_binCRTT_muon2018_bin3
396 prop_binCRF_electron2017_bin9_WrongSign
397 prop_binCRTT_electron2018_bin9
398 lep_2017
399 prop_binCRTT_muon2016M_bin13
400 prop_binCRTT_electron2017_bin13
401 prop_binCRTT_electron2016M_bin2
402 prop_binCROS_muon2016M_bin11
403 prop_binCRF_muon2016M_bin11
404 prop_binCROS_muon2016M_bin12
405 prop_binCRTT_muon2017_bin12
406 prop_binCROS_electron2016M_bin12_WrongSign
407 prop_binCRTT_electron2018_bin3
408 prop_binCROS_electron2016M_bin11
409 prop_binCRTT_electron2016M_bin13
410 prop_binCRTT_electron2018_bin10
411 prop_binCRTT_electron2016M_bin9
412 prop_binCRTT_electron2017_bin7
413 lep_2018
414 prop_binCRF_electron2017_bin8_Triboson
415 prop_binSR_electron2016M_bin13_ZZtoLep
416 prop_binCRF_muon2016M_bin12
417 prop_binCRTT_electron2018_bin11
418 prop_binCROS_muon2016M_bin5_TTTo2L2Nu
419 prop_binSR_electron2016M_bin1_WZ
420 prop_binSR_electron2016M_bin1_TTTo2L2Nu
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



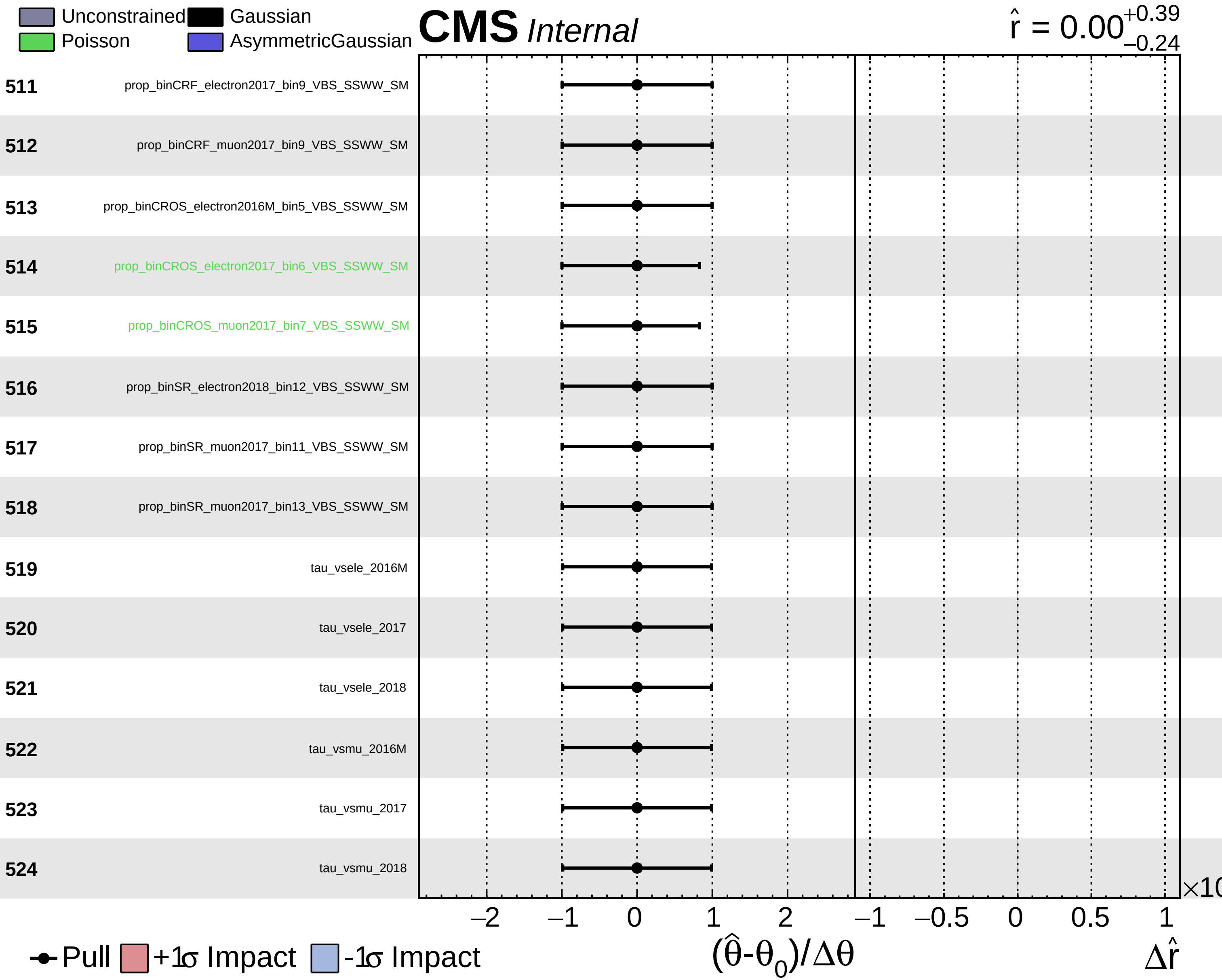
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
421 lep_2016M
422 prop_binCRTT_electron2017_bin11
423 prop_binCRTT_electron2018_bin4
424 prop_binCRF_electron2017_bin8_ZZtoLep
425 prop_binCRTT_electron2017_bin4
426 prop_binCRF_electron2017_bin9_WpWpJJ_QCD
427 prop_binCROS_muon2016M_bin10
428 prop_binCRTT_electron2017_bin8
429 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
430 prop_binSR_electron2016M_bin1_VG
431 prop_binCROS_electron2018_bin11_TTTo2L2Nu
432 prop_binCROS_muon2017_bin7_TTTo2L2Nu
433 prop_binCROS_muon2017_bin11
434 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
435 prop_binCROS_electron2017_bin11
436 prop_binCRF_electron2016M_bin7_ZZtoLep
437 prop_binSR_electron2016M_bin13_WrongSign
438 prop_binCRF_muon2018_bin13
439 prop_binCRF_electron2017_bin9_Triboson
440 prop_binCROS_muon2017_bin13
441 prop_binCRF_muon2017_bin9_WrongSign
442 prop_binCRTT_electron2018_bin7
443 prop_binSR_electron2016M_bin1_ZZtoLep
444 prop_binCROS_electron2017_bin7_TTTo2L2Nu
445 prop_binCROS_electron2017_bin13
446 prop_binCROS_electron2016M_bin10
447 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
448 prop_binCROS_electron2016M_bin7_Fake
449 prop_binCROS_electron2016M_bin12_Fake
450 prop_binCRF_electron2016M_bin7_TVX
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
451 prop_binCRF_muon2017_bin9_TVX
452 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
453 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
454 prop_binCROS_electron2018_bin11_TVX
455 prop_binCRF_muon2017_bin9_Triboson
456 prop_binSR_electron2016M_bin1_TVX
457 prop_binCROS_electron2017_bin6_TTTo2L2Nu
458 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
459 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
460 prop_binCROS_electron2017_bin7_TVX
461 prop_binSR_electron2016M_bin1_Triboson
462 prop_binCROS_electron2018_bin11_Fake
463 prop_binCROS_electron2016M_bin5_VG
464 prop_binCRF_electron2016M_bin7_Triboson
465 prop_binCRF_electron2017_bin13
466 prop_binCRF_electron2016M_bin7_WpWpJJ_QCD
467 prop_binCROS_electron2018_bin11_Triboson
468 FES
469 prop_binCROS_electron2016M_bin12_Triboson
470 prop_binCROS_electron2016M_bin7_Triboson
471 prop_binCROS_muon2016M_bin5_VG
472 prop_binCROS_muon2017_bin7_TVX
473 prop_binCROS_electron2016M_bin12_TVX
474 prop_binCROS_electron2018_bin11_ZZtoLep
475 prop_binCROS_electron2016M_bin7_TVX
476 prop_binCROS_muon2017_bin7_ZZtoLep
477 prop_binCROS_muon2017_bin7_Triboson
478 prop_binCROS_electron2016M_bin5_ZZtoLep
479 prop_binCROS_electron2017_bin7_Triboson
480 prop_binCROS_electron2016M_bin7_ZZtoLep
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
481 prop_binCROS_electron2017_bin7_ZZtoLep
482 prop_binCROS_electron2016M_bin12_ZZtoLep
483 prop_binCROS_electron2017_bin6_ZZtoLep
484 prop_binCROS_electron2016M_bin5_TVX
485 prop_binCROS_electron2016M_bin5_Triboson
486 prop_binCROS_electron2018_bin11_VBS_SSWW_SM
487 FSR_sig
488 prop_binCROS_muon2016M_bin5_Triboson
489 prop_binCROS_electron2017_bin6_TVX
490 prop_binCROS_muon2016M_bin5_TVX
491 prop_binCROS_muon2016M_bin5_ZZtoLep
492 prop_binCROS_electron2017_bin6_Triboson
493 ISR_sig
494 prop_binSR_electron2017_bin11_VBS_SSWW_SM
495 VBS_2018
496 QCDScale_sig
497 prop_binCRF_electron2016M_bin11_VBS_SSWW_SM
498 prop_binCROS_muon2016M_bin5_VBS_SSWW_SM
499 VBS_2017
500 prop_binSR_electron2016M_bin13_VBS_SSWW_SM
501 prop_binSR_muon2016M_bin10_VBS_SSWW_SM
502 prop_binCRF_electron2016M_bin7_VBS_SSWW_SM
503 prop_binCRF_electron2017_bin8_VBS_SSWW_SM
504 prop_binSR_electron2016M_bin1_VBS_SSWW_SM
505 prop_binCROS_electron2016M_bin12_VBS_SSWW_SM
506 VBS_2016M
507 prop_binCROS_electron2017_bin7_VBS_SSWW_SM
508 prop_binCROS_electron2016M_bin7_VBS_SSWW_SM
509 prop_binSR_muon2016M_bin13_VBS_SSWW_SM
510 QCDScale_WZ
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.39}_{-0.24}$
511 prop_binCRF_electron2017_bin9_VBS_SSWW_SM
512 prop_binCRF_muon2017_bin9_VBS_SSWW_SM
513 prop_binCROS_electron2016M_bin5_VBS_SSWW_SM
514 prop_binCROS_electron2017_bin6_VBS_SSWW_SM
515 prop_binCROS_muon2017_bin7_VBS_SSWW_SM
516 prop_binSR_electron2018_bin12_VBS_SSWW_SM
517 prop_binSR_muon2017_bin11_VBS_SSWW_SM
518 prop_binSR_muon2017_bin13_VBS_SSWW_SM
519 tau_vsele_2016M
520 tau_vsele_2017
521 tau_vsele_2018
522 tau_vsmu_2016M
523 tau_vsmu_2017
524 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$